Online Supplement: Carbon-Aware Scheduling for Phase-Disaggregated LLM Inference in Cloud and Serverless Platforms

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Revision v2 – Artifact Supplement

Code and Reproducibility Repository: https://github.com/UmarIqbal000/carbon-aware-llm-scheduling (SimPy 4.1.2, paired seeds 100-109, runner experiments/reproduce_all.py).

S1. PRISMA-2020 V2 Corpus Reconciliation Summary

Table 1 summarizes the updated V2 screening accounting logged in research/search_log.md, expanding the V1 corpus ($n = 47$) with 27 newly screened V2 candidate records (13 explicitly audited anchors including the four late-2025/2026 P/D and geo-routing works plus 14 additional 2024–Nov 2026 search hits), removing 2 duplicates, excluding 3 items at title/abstract and 4 items at full-text eligibility, and retaining $n = 65$ unique works.

Table 1: Updated V2 Corpus Characteristics ($n = 65$ Included Works)

Dimension	**Verified Count and Composition**
Screening Flow	102 identified → 3 duplicates removed → 99 screened → 14 title/abstract excluded → 85 assessed → 20 full-text excluded → **65 included**.
Source Channels	Channel 1 (surveys): 9; Channel 2 (bounded API queries): 14; Channel 3 (targeted anchors, benchmarks, standards): 42 ($9 + 14 + 42 = 65$).
Thematic Pillars	Pillar (a) surveys/agendas: 11; Pillar (b) carbon cloud/serverless/geo: 18; Pillar (c) P/D & LLM serving: 21; Pillar (d) embodied/traces/standards: 13; Pillar (e) foundations: 2.
Publication Recency	60/65 from 2022–2026 (**92.3%**); 38/65 from 2025–2026 (**58.5%**); exact counts: 2011 (2), 2020 (1), 2021 (2), 2022 (1), 2023 (6), 2024 (15), 2025 (16), 2026 (22).
Audit Depth	60/65 full-text audited (**92.3%**); 5/65 abstract-only (**7.7%**).
Preprint Status	18 preprints total: 17 arXiv-only preprints plus 1 accepted ATC'26 conference preprint (wang2026festina, marked % TODO-VERIFY after Crossref DOI lookup returned HTTP 404).

S2. Calibration Anchors vs. Held-Out Benchmark Validation (24 Checks)

All 24 quantitative checks generated by experiments/validate_simulator.py (saved in validation_benchmarks.csv) appear in Table 2, explicitly separated into **19 calibration anchors**

(fitted directly from Splitwise Table IV, Figs. 8–9 power-cap curves, and ServerlessLLM Fig. 6a checkpoint loader timings) and **5 held-out benchmark sweeps** (comparing simulated `B2` vs. `B1` energy-per-token across 0.25×–2.00× RPS against Li et al.'s empirical trends, which were not used during parameter fitting). Across uncontended single-request Splitwise Table IV points, simulated TTFT includes parameterized KV stall overhead (+5.0 ms on H100, +8.0 ms on A100), yielding a +8.81% to +14.24% positive offset above raw Splitwise compute latencies, while nominal 28/42 ms decode step calibration differs from code-workload reference TPOT (−9.68% H100, −19.23% A100). Under power capping, a 0.55× TDP decode cap leaves 16-request decode step latency flat at 29.25 ms while dropping node power from 1563.5 W to 1069.8 W (−31.6%), before an abrupt +6.0% step penalty appears at 0.40× TDP (31.01 ms), matching Splitwise Figs. 8–9.

Table 2: Calibration Anchors (19) and Held-Out Benchmark Checks (5) in `validation_benchmarks.csv`

Type	Suite	Condition	Sim	Ref	Err (%)	Notes
Calib. anchor	Splitwise Table IV (Conv)	H100 ($L_{\text{in}} = 1113$), TTFT (ms)	95.96	84.0	+14.24	raw=84.0 + kv_stall=5.0 ms
Calib. anchor	Splitwise Table IV (Conv)	H100 ($L_{\text{in}} = 1113$), TPOT (ms)	28.00	28.0	0.00	0.55 TDP preserves TBT
Calib. anchor	Splitwise Table IV (Conv)	A100 ($L_{\text{in}} = 1083$), TTFT (ms)	169.73	155.0	+9.50	raw=155.0 + kv_stall=8.0 ms
Calib. anchor	Splitwise Table IV (Conv)	A100 ($L_{\text{in}} = 1083$), TPOT (ms)	42.00	40.0	+5.00	0.55 TDP preserves TBT
Calib. anchor	Splitwise Table IV (Code)	H100 ($L_{\text{in}} = 1290$), TTFT (ms)	108.04	95.0	+13.73	raw=95.0 + kv_stall=5.0 ms
Calib. anchor	Splitwise Table IV (Code)	H100 ($L_{\text{in}} = 1290$), TPOT (ms)	28.00	31.0	-9.68	0.55 TDP preserves TBT
Calib. anchor	Splitwise Table IV (Code)	A100 ($L_{\text{in}} = 1333$), TTFT (ms)	201.29	185.0	+8.81	raw=185.0 + kv_stall=8.0 ms
Calib. anchor	Splitwise Table IV (Code)	A100 ($L_{\text{in}} = 1333$), TPOT (ms)	42.00	52.0	-19.23	0.55 TDP preserves TBT
Calib. anchor	Splitwise Figs 8–9	H100 batch-16, cap=1.00 TDP	29.25	28.0	+4.47	Power = 1563.5 W
Calib. anchor	Splitwise Figs 8–9	H100 batch-16, cap=0.70 TDP	29.25	28.0	+4.47	Power = 1234.4 W
Calib. anchor	Splitwise Figs 8–9	H100 batch-16, cap=0.55 TDP	29.25	28.0	+4.47	Power = 1069.8 W
Calib. anchor	Splitwise Figs 8–9	H100 batch-16, cap=0.50 TDP	29.25	28.0	+4.47	Power = 1015.0 W
Calib. anchor	Splitwise Figs 8–9	H100 batch-16, cap=0.40 TDP	31.01	–	–	Power = 905.3 W (+6.0% step)
Calib. anchor	ServerlessLLM Fig 6a	Llama-2-70B SLLM / Safe / PyT (s)	10.3 / 48.0 / 84.0	10.3 / 48.0 / 84.0	0.00	cold=2183.5 W, idle=724.6 W
Calib. anchor	ServerlessLLM Fig 6a	OPT-13B SLLM / Safe / PyT (s)	2.0 / 8.2 / 14.0	2.0 / 8.2 / 14.0	0.00	cold=545.9 W, idle=181.2 W
Held-out	Li et al. (2026) Sweep	Conv 1h 0.25x RPS (B2 vs B1 J/tok)	2.195	2.745	-20.01	TTFT +11.4%, TPOT -7.8%
Held-out	Li et al. (2026) Sweep	Conv 1h 0.50x RPS (B2 vs B1 J/tok)	1.138	1.509	-24.57	TTFT +11.5%, TPOT -14.9%
Held-out	`Li2026` (1.00x RPS)	Conv 1h 1.00x RPS (B2 vs B1 J/tok)	0.813	1.163	-30.10	TTFT +11.7%, TPOT -19.4%
Held-out	`Li2026` (1.50x RPS)	Conv 1h 1.50x RPS (B2 vs B1 J/tok)	0.628	0.903	-30.40	TTFT +25.7%, TPOT -26.2%
Held-out	`Li2026` (2.00x RPS)	Conv 1h 2.00x RPS (B2 vs B1 J/tok)	0.567	0.810	-30.04	TTFT +44.3%, TPOT -31.6%

S3. Complete 24-Hour Trace-Tiling Sensitivity Ablation

Table 3 reports all 12 policy-by-tiling combinations across four synthetic 24-hour trace construction modes (10 paired seeds each = 120 runs total) from `tiling_ablation_summary.csv`. Across all four regimes, `P4` operational SCI remains bounded within 0.0314–0.0320 g/1k tok (maximum relative shift −2.02%) and SLO violation rates remain within 0.817%–0.835%, confirming that bootstrap block reuse does not artificially create the 60.7% carbon advantage over `B1`.

Table 3: Full 24h Trace-Tiling Sensitivity Matrix (10 Paired Seeds per Cell)

Tiling Mode	Policy	SCI_{op} (g/1k)	SCI_{tot} (g/1k)	SLO (%)	P95 TTFT (ms)	Mean Reqs
`orig_280s_repl`	`B1`	0.0815 ± 0.0097	0.0906 ± 0.0097	0.0004 ± 0.0003	268.6	476,119.6
`orig_280s_repl`	`B2`	0.0573 ± 0.0068	0.0663 ± 0.0068	0.1996 ± 0.0157	300.3	476,119.6
`orig_280s_repl`	`P4`	0.0320 ± 0.0061	0.0411 ± 0.0061	0.8237 ± 0.0386	504.5	476,119.6
`perm_280s_norepl`	`B1`	0.0806 ± 0.0094	0.0896 ± 0.0094	0.0008 ± 0.0003	268.7	478,841.8
`perm_280s_norepl`	`B2`	0.0567 ± 0.0066	0.0656 ± 0.0066	0.2164 ± 0.0086	300.6	478,841.8
`perm_280s_norepl`	`P4`	0.0316 ± 0.0059	0.0406 ± 0.0059	0.8174 ± 0.0291	511.2	478,841.8
`fine_90s_perm`	`B1`	0.0807 ± 0.0094	0.0896 ± 0.0094	0.0008 ± 0.0005	268.6	478,556.8
`fine_90s_perm`	`B2`	0.0567 ± 0.0066	0.0657 ± 0.0066	0.2117 ± 0.0094	300.6	478,556.8
`fine_90s_perm`	`P4`	0.0317 ± 0.0059	0.0406 ± 0.0059	0.8202 ± 0.0333	510.9	478,556.8
`coarse_900s_perm`	`B1`	0.0801 ± 0.0091	0.0890 ± 0.0091	0.0008 ± 0.0003	268.7	482,902.5
`coarse_900s_perm`	`B2`	0.0564 ± 0.0065	0.0652 ± 0.0064	0.2240 ± 0.0098	300.9	482,902.5
`coarse_900s_perm`	`P4`	0.0314 ± 0.0057	0.0402 ± 0.0057	0.8351 ± 0.0278	514.9	482,902.5

S4. Path C Cross-Region Activation Regime (SW5, 690 Runs)

Table 4 reports the mean Path C share (%) across 10 paired seeds ('100–109') as WAN bandwidth (10–400 Gbps), WAN RTT (5–200 ms), carbon differential, and request load vary under P4. Path C stays identically 0.000% whenever BW $\leq$ 25 Gbps and RTT $\geq$ 20 ms, and peaks at 0.308% under congested 2.0× conv with 400 Gbps / 5 ms links.

Table 4: SW5 Cross-Region Path C Share (%) by WAN Bandwidth and RTT ($N = 10$ Seeds)

Regime (Policy P4)	BW (Gbps)	5 ms	20 ms	50 ms	100 ms	200 ms
conv 2.0×, High ΔCI (S.Scot-TEPCO)	400	0.308	0.296	0.221	0.074	0.027
conv 2.0×, High ΔCI (S.Scot-TEPCO)	100	0.286	0.276	0.103	0.042	0.000
conv 2.0×, High ΔCI (S.Scot-TEPCO)	25	0.032	0.000	0.000	0.000	0.000
conv 2.0×, High ΔCI (S.Scot-TEPCO)	10	0.007	0.000	0.000	0.000	0.000
conv 2.0×, Low ΔCI (S.Scot-NW Eng)	400	0.228	0.227	0.177	0.041	0.012
conv 2.0×, Low ΔCI (S.Scot-NW Eng)	100	0.204	0.201	0.072	0.023	0.000
conv 2.0×, Low ΔCI (S.Scot-NW Eng)	25	0.027	0.000	0.000	0.000	0.000
conv 2.0×, Low ΔCI (S.Scot-NW Eng)	10	0.003	0.000	0.000	0.000	0.000
conv 1.0×, High ΔCI (S.Scot-TEPCO)	400 / 100	0.060 / 0.059	0.046 / 0.046	0.033 / 0.011	0.000 / 0.000	0.000 / 0.000
conv 1.0×, High ΔCI (S.Scot-TEPCO)	25 / 10	0.007 / 0.001	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000
code 2.0×, Med ΔCI (S.Scot-London)	400 / 100 / 10	0.004 / 0.003 / 0.000	0.009 / 0.007 / 0.000	–	0.003 / 0.001 / 0.000	–

S5. Clean-Region Scale-Out Break-Even Curve (SW6, 560 Runs)

Table 5 summarizes all 56 configurations (10 paired seeds each) sweeping demand from 1.0× to 3.0× RPS across 6-node equal clusters (P3_equal, P4_equal) vs. +4 clean South Scotland nodes (P3+4, P4+4).

Table 5: SW6 Total Lifecycle SCI (g/1k) and SLO Violations Across Demand Scale ($N = 10$ Seeds)

Lifetime	RPS	P4_eq (SLO)	P4+4 (SLO)	Δ(P4+4 vs. P4_eq)	P3_eq (SLO)	P3+4 (SLO)
4 yr uniform	1.00×	0.01735 (0.08%)	0.02301 (0.02%)	+32.66%	0.01719 (11.54%)	0.02300 (0.54%)
4 yr uniform	1.25×	0.01442 (0.10%)	0.01831 (0.02%)	+26.96%	0.01400 (20.93%)	0.01830 (1.44%)
4 yr uniform	1.50×	0.01299 (0.13%)	0.01476 (0.04%)	+13.62%	0.01215 (30.92%)	0.01485 (2.16%)
4 yr uniform	1.75×	0.01344 (0.14%)	0.01310 (0.05%)	−2.59%	0.01190 (37.98%)	0.01333 (3.03%)
4 yr uniform	2.00×	0.01374 (0.16%)	0.01187 (0.07%)	−13.61%	0.01174 (44.81%)	0.01221 (5.12%)
4 yr uniform	2.50×	0.01299 (0.36%)	0.00955 (0.14%)	−26.49%	0.01089 (54.25%)	0.00983 (13.10%)
4 yr uniform	3.00×	0.01292 (1.17%)	0.00896 (0.20%)	−30.63%	0.01069 (67.12%)	0.00923 (22.50%)
9 yr host / 3 yr GPU	1.00×	0.01516 (0.08%)	0.01937 (0.02%)	+27.76%	0.01501 (11.54%)	0.01937 (0.54%)
9 yr host / 3 yr GPU	1.25×	0.01270 (0.10%)	0.01543 (0.02%)	+21.56%	0.01227 (20.93%)	0.01542 (1.44%)
9 yr host / 3 yr GPU	1.50×	0.01160 (0.13%)	0.01245 (0.04%)	+7.30%	0.01074 (30.92%)	0.01253 (2.16%)
9 yr host / 3 yr GPU	1.75×	0.01222 (0.14%)	0.01106 (0.05%)	−9.51%	0.01061 (37.98%)	0.01127 (3.03%)
9 yr host / 3 yr GPU	2.00×	0.01265 (0.16%)	0.01005 (0.07%)	−20.55%	0.01054 (44.81%)	0.01034 (5.12%)
9 yr host / 3 yr GPU	2.50×	0.01215 (0.36%)	0.00814 (0.14%)	−32.99%	0.00991 (54.25%)	0.00836 (13.10%)
9 yr host / 3 yr GPU	3.00×	0.01219 (1.17%)	0.00775 (0.20%)	−36.45%	0.00982 (67.12%)	0.00791 (22.50%)

S6. Explicit Simulator Calibration Assumptions

Table 6 documents non-measured modeling parameters and engineering assumptions fixed across the main experiments and varied in sensitivity sweeps SW1-SW6.

Table 6: Explicit Simulator Calibration Assumptions (Moved from Main Text Sec. V-B)

Mechanism	Baseline Value	Sensitivity Range
Synthetic WAN RTT	0.5 / 20 / 30 / 200 ms	Varied 5–200 ms (SW5)
WAN KV Bandwidth	10 Gbps	10, 25, 100, 400 Gbps (SW3c, SW5)
Sublinear MFU γ	0.70 ($\text{mfu}_{\text{sat}} = 0.45$)	Physical saturation envelope
Cold Power Fraction	0.65 P_{TDP}	Swept in $[0.45, 0.85]$ (SW4)
24h Trace Extension	Empirical block tiling	Verified in Table 3

S7. WAN Bandwidth Sweep (SW3c) Path Mix

Table 7 reports the full 10-row routing path mix and tail latency breakdown for the WAN bandwidth sweep (SW3c).

Table 7: WAN Bandwidth Sweep (SW3c) Path Mix and Performance ($N = 10$ Seeds)

Work.	BW	Pol.	SCI_{op}	SLO	P95 (ms)	A	B	C	D (%)
code	10G	P3	0.04441	44.45%	39,318	0.0	56.6	43.4	0.0
code	10G	P4	0.04419	**6.57%**	**1,614**	14.1	39.8	**0.0**	**46.1**
code	25G	P3	0.04403	32.12%	7,110	0.0	56.7	43.3	0.0
code	100G	P3	0.04391	20.78%	3,341	0.0	56.7	43.3	0.0
code	100G	P4	0.04419	**6.57%**	**1,614**	14.1	39.8	**0.0**	**46.1**
conv 2x	10G	P3	0.02481	42.20%	162,595	0.0	49.8	50.2	0.0
conv 2x	10G	P4	0.03191	**1.10%**	**881**	1.6	35.5	**0.0**	**63.0**
conv 2x	25G	P3	0.02502	9.63%	1,694	0.0	49.1	50.9	0.0
conv 2x	100G	P3	0.02488	3.76%	1,492	0.0	49.5	50.5	0.0
conv 2x	100G	P4	0.03193	**1.10%**	**881**	1.6	35.4	**0.02**	**63.0**